Numpy Intro_05-03-2026

In [1]:
```
import numpy as np
```

In [2]:
```
np.__version__
```

Out[2]:
```
'2.3.5'
```

In [3]:
```
my_list = [0,1,2,3,4,5]
my_list
```

Out[3]:
```
[0, 1, 2, 3, 4, 5]
```

In [4]:
```
type(my_list)
```

Out[4]:
```
list
```

In [5]:
```
arr = np.array(my_list)
arr
```

Out[5]:
```
array([0, 1, 2, 3, 4, 5])
```

In [6]:
```
type(arr)
```

Out[6]:
```
numpy.ndarray
```

In [7]:
```
np.arange(10)  #arange it will print the value within interval
```

Out[7]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [8]:
```
np.arange(10,15)
```

Out[8]:
```
array([10, 11, 12, 13, 14])
```

In [9]:
```
np.arange(10,20,5)
```

Out[9]:
```
array([10, 15])
```

In [10]:
```
np.arange(10,20,5,2)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[10], line 1
----> 1 np.arange(10,20,5,2)

TypeError: Cannot interpret '2' as a data type
```

In [11]:
```
np.zeros(5) #parameter tunning
```

Out[11]:
```
array([0., 0., 0., 0., 0.])
```

In [12]:
```
np.zeros(5, dtype=int) #hyperparameter tunning
```

Out[12]:
```
array([0, 0, 0, 0, 0])
```

In [13]:
```
np.zeros([2,2], dtype = int) #2d array
```

Out[13]:
```
array([[0, 0],
       [0, 0]])
```

In [14]:
```
np.zeros([5,10], dtype = int)
```

Out[14]:
```
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

In [15]:
```
np.ones(3)
```

Out[15]:
```
array([1., 1., 1.])
```

In [16]:
```
np.ones([3,5], dtype = int)
```

Out[16]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [17]:
```
np.ones([3,5])
```

Out[17]:
```
array([[1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.]])
```

In [18]:
```
np.arange(10,20)
```

Out[18]:
```
array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

In [19]:
```
np.arange(20, 10)
```

Out[19]:
```
array([], dtype=int64)
```

In [20]:
```
np.arange(-20, 10)
```

Out[20]:
```
array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10,  -9,  -8,
        -7,  -6,  -5,  -4,  -3,  -2,  -1,   0,   1,   2,   3,   4,   5,
         6,   7,   8,   9])
```

In [21]:
```
np.twos(3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[21], line 1
----> 1 np.twos(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

In [22]:
```
np.threes(3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[22], line 1
----> 1 np.threes(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'threes'
```

In [23]:
```
random.rand(3,2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[23], line 1
----> 1 random.rand(3,2)

NameError: name 'random' is not defined
```

In [24]:
```
np.random.rand(2)
```

Out[24]:
```
array([0.51985352, 0.27230226])
```

In [25]:
```
np.random.rand(5)
```

Out[25]:
```
array([0.41108553, 0.60636591, 0.67580983, 0.34002852, 0.68584988])
```

In [26]:
```
np.random.rand(3,2)
```

Out[26]:
```
array([[0.40619343, 0.91390489],
       [0.28143111, 0.70575157],
       [0.1770155 , 0.27051321]])
```

In [27]:
```
np.random.randint(3)
```

Out[27]:
```
0
```

In [28]:
```
np.random.randint(5,10)
```

Out[28]:
```
7
```

In [29]:
```
np.random.randint(5,10,3)
```

Out[29]:
```
array([8, 9, 6], dtype=int32)
```

In [30]:
```
ran = np.random.randint(5,10,3)
print(ran)
```

```
[6 9 6]
```

In [31]:
```
otp = np.random.randint(5,10,4)
print(otp)
```

```
[7 6 6 6]
```

In [32]:
```
arr0 = np.random.randint(10,40,(10,10))
print(arr0)
```

```
[[19 33 11 18 38 37 11 17 32 35]
 [31 18 21 26 13 18 17 28 35 29]
 [34 25 17 36 12 35 29 25 15 33]
 [21 35 34 32 15 13 22 11 36 22]
 [22 17 22 24 27 22 35 28 11 33]
 [22 38 22 20 23 32 20 11 16 24]
 [26 21 28 21 19 37 29 37 33 20]
 [33 37 31 31 21 32 26 27 36 39]
 [18 35 32 14 31 34 21 30 27 28]
 [35 18 19 36 25 24 28 34 25 16]]
```

In [33]:
```
arr0
```

Out[33]:
```
array([[19, 33, 11, 18, 38, 37, 11, 17, 32, 35],
       [31, 18, 21, 26, 13, 18, 17, 28, 35, 29],
       [34, 25, 17, 36, 12, 35, 29, 25, 15, 33],
       [21, 35, 34, 32, 15, 13, 22, 11, 36, 22],
       [22, 17, 22, 24, 27, 22, 35, 28, 11, 33],
       [22, 38, 22, 20, 23, 32, 20, 11, 16, 24],
       [26, 21, 28, 21, 19, 37, 29, 37, 33, 20],
       [33, 37, 31, 31, 21, 32, 26, 27, 36, 39],
       [18, 35, 32, 14, 31, 34, 21, 30, 27, 28],
       [35, 18, 19, 36, 25, 24, 28, 34, 25, 16]], dtype=int32)
```

In [34]:
```
arr1 = np.random.randint(1,5,(3,3))
print(arr1)
```

```
[[3 1 3]
 [1 4 4]
 [4 4 2]]
```

, indexing, slicing in array

In [35]:
```
arr1[:]
```

Out[35]:
```
array([[3, 1, 3],
       [1, 4, 4],
       [4, 4, 2]], dtype=int32)
```

In [36]:
```
len(arr1) #
```

Out[36]:
```
3
```

In [37]:
```
arr1.shape
```

Out[37]:
```
(3, 3)
```

In [38]:
```
arr0.shape
```

Out[38]:
```
(10, 10)
```

In [39]:
```
len(arr0)
```

Out[39]:
```
10
```

In [40]:
```
arr1
```

Out[40]:
```
array([[3, 1, 3],
       [1, 4, 4],
       [4, 4, 2]], dtype=int32)
```

In [41]:
```
arr1[0] # print row
```

Out[41]:
```
array([3, 1, 3], dtype=int32)
```

In [42]:
```
arr1[1]  # print row
```

Out[42]:
```
array([1, 4, 4], dtype=int32)
```

In [43]:
```
arr1
```

Out[43]:
```
array([[3, 1, 3],
       [1, 4, 4],
       [4, 4, 2]], dtype=int32)
```

In [44]:
```
arr1[:,1]
```

Out[44]:
```
array([1, 4, 4], dtype=int32)
```

In [45]:
```
arr1[:,2]
```

Out[45]:
```
array([3, 4, 2], dtype=int32)
```

In [46]:
```
arr1
```

Out[46]:
```
array([[3, 1, 3],
       [1, 4, 4],
       [4, 4, 2]], dtype=int32)
```

In [47]:
```
arr1[0]
```

Out[47]:
```
array([3, 1, 3], dtype=int32)
```

In [48]:
```
v = arr1[1, 2]
print(v)
```

```
4
```

In [49]:
```
arr0
```

Out[49]:
```
array([[19, 33, 11, 18, 38, 37, 11, 17, 32, 35],
       [31, 18, 21, 26, 13, 18, 17, 28, 35, 29],
       [34, 25, 17, 36, 12, 35, 29, 25, 15, 33],
       [21, 35, 34, 32, 15, 13, 22, 11, 36, 22],
       [22, 17, 22, 24, 27, 22, 35, 28, 11, 33],
       [22, 38, 22, 20, 23, 32, 20, 11, 16, 24],
       [26, 21, 28, 21, 19, 37, 29, 37, 33, 20],
       [33, 37, 31, 31, 21, 32, 26, 27, 36, 39],
       [18, 35, 32, 14, 31, 34, 21, 30, 27, 28],
       [35, 18, 19, 36, 25, 24, 28, 34, 25, 16]], dtype=int32)
```

In [50]:
```
arr0[-7]
```

Out[50]:
```
array([21, 35, 34, 32, 15, 13, 22, 11, 36, 22], dtype=int32)
```

In [51]:
```
arr0[:, -7]
```

Out[51]:
```
array([18, 26, 36, 32, 24, 20, 21, 31, 14, 36], dtype=int32)
```

In [52]:
```
arr0
```

Out[52]:
```
array([[19, 33, 11, 18, 38, 37, 11, 17, 32, 35],
       [31, 18, 21, 26, 13, 18, 17, 28, 35, 29],
       [34, 25, 17, 36, 12, 35, 29, 25, 15, 33],
       [21, 35, 34, 32, 15, 13, 22, 11, 36, 22],
       [22, 17, 22, 24, 27, 22, 35, 28, 11, 33],
       [22, 38, 22, 20, 23, 32, 20, 11, 16, 24],
       [26, 21, 28, 21, 19, 37, 29, 37, 33, 20],
       [33, 37, 31, 31, 21, 32, 26, 27, 36, 39],
       [18, 35, 32, 14, 31, 34, 21, 30, 27, 28],
       [35, 18, 19, 36, 25, 24, 28, 34, 25, 16]], dtype=int32)
```

In [53]:
```
arr0[3:4, 2:6]
```

Out[53]:
```
array([[34, 32, 15, 13]], dtype=int32)
```

In [54]:
```
arr1
```

Out[54]:
```
array([[3, 1, 3],
       [1, 4, 4],
       [4, 4, 2]], dtype=int32)
```

In [55]:
```
arr1[1]
```

Out[55]:
```
array([1, 4, 4], dtype=int32)
```

In [56]:
```
arr1[1,2] # 1 first row & 2 means colums
```

Out[56]:
```
np.int32(4)
```

In [57]:
```
arr1
```

Out[57]:
```
array([[3, 1, 3],
       [1, 4, 4],
       [4, 4, 2]], dtype=int32)
```

In [58]:
```
arr1[1:2]
```

Out[58]:
```
array([[1, 4, 4]], dtype=int32)
```